

MindScope: A Cloud-Native Multimodal Depression Severity Assessment Framework Using Audio, Video, Text, and Behavioral Intelligence

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Abstract—We present MindScope, a cloud-native multimodal framework for depression severity assessment combining audio, facial-video, and transcript modalities with behavioral intelligence. MindScope integrates transformer-based cross-modal fusion with Dynamic Modality Gating (DMG), Missing Modality Recovery (MMR), and a Confidence-Aware Reliability Index (CARI) to provide calibrated predictions and clinician-facing explanations. The architecture is designed for production deployment: a FastAPI orchestration layer, stateless PyTorch inference service, distributed caching, and support for federated updates. We evaluate MindScope on the DAIC-WOZ benchmark using a subject-aware split and demonstrate consistent gains over unimodal baselines: relative improvements of 12–18

Index Terms—Multimodal Depression Assessment, Affective Computing, Behavioral Intelligence, Explainable AI, Transformer Fusion, Mental Health AI, Audio-Visual Analytics, Federated Learning, Clinical AI Systems

I. INTRODUCTION

Depression is a leading cause of disability worldwide and often remains under-assessed in routine care. Standardised questionnaires such as the PHQ-series provide useful screening signals but rely on self-report and periodic clinician contact, creating barriers to continuous monitoring. Automated approaches using speech, facial behaviour, and language offer complementary signals, yet prior work has often focused on model performance in isolation rather than operational readiness.

This paper presents MindScope, a multimodal depression severity assessment framework that explicitly targets both algorithmic performance and deployment realism. MindScope fuses acoustic, visual, and semantic signals with a transformer-based fusion core; it uses Dynamic Modality Gating (DMG) to adapt modality contributions, Missing Modality Recovery (MMR) to synthesize surrogate representations when a modality is absent, and the Confidence-Aware Reliability Index (CARI) to quantify prediction reliability for clinical review.

Major Contributions of this Paper:

- We propose a production-oriented multimodal architecture combining transformer fusion, DMG, MMR, and CARI for depression severity assessment.
- We provide a reproducible evaluation on DAIC-WOZ with detailed ablations quantifying the impact of each module under realistic modality dropout and noise.

- We introduce practical deployment guidance including streaming inference patterns, caching strategies, and latency/throughput measurements for clinical workflows.
- We demonstrate explainability techniques that expose modality contributions and attention-based attributions to clinicians.
- We open a pathway to federated model updates to preserve site-level privacy while improving generalization.

II. RELATED WORK

Research in automated depression assessment spans unimodal and multimodal approaches. Speech-based systems exploit prosodic, spectral, and voice-quality cues [1], [2]. Facial-affect analysis via OpenFace and deep CNNs extracts action units and micro-expression patterns relevant to affective states [3]. Language models and sentence embeddings provide semantic context via SBERT and transformer encoders [4].

Multimodal fusion strategies increasingly rely on attention and transformer-based architectures that learn cross-modal interactions [5]. Robustness to missing modalities is typically addressed with imputation or gating mechanisms; our DMG and MMR build on these ideas and prioritise operational constraints such as bounded modality weights and explicit confidence penalties [6]. Explainability in clinical AI focuses on feature attributions and calibrated confidence to aid human review [7]. Federated learning enables privacy-preserving updates across institutions and has seen growing interest for healthcare AI [8].

Despite these advances, gaps remain: limited operational validation, few systems combine explainability with reliability estimation, and deployment-oriented evaluations (latency, streaming) are sparse. MindScope addresses these gaps by uniting algorithmic components with deployment-aware engineering.

III. PROPOSED METHODOLOGY

We formalise an input session as a time-aligned set of modality streams: audio waveform $A(t)$, video frames $V(t)$, and transcript segments $T(t)$. The goal is to predict a continuous PHQ-derived severity score $y \in \mathbb{R}$ and an associated reliability score $r \in [0, 1]$.

A. Modality Encoders

Each modality is encoded into a sequence of tokens at a common frame rate (e.g., 100 ms frames):

$$\mathbf{a}_i = E_{\text{audio}}(A[t_i : t_{i+1}]) \quad (1)$$

$$\mathbf{v}_i = E_{\text{video}}(V[t_i]) \quad (2)$$

$$\mathbf{t}_i = E_{\text{text}}(T[t_i]) \quad (3)$$

Audio features combine 39-D MFCC + Δ + $\Delta\Delta$, eGeMAPS descriptors, and prosodic statistics. Video uses OpenFace-derived action units, gaze and head-pose vectors, complemented by framewise CNN embeddings. Text uses Whisper ASR output followed by Sentence-BERT (SBERT) embeddings projected to the shared latent space.

B. Temporal Alignment and Projection

Tokens are projected to a shared d -dimensional space via modality-specific linear projections: $\tilde{\mathbf{a}}_i = W_a \mathbf{a}_i$, etc. We align tokens by nearest-frame matching and temporal interpolation to create a sequence $\{\mathbf{x}_i\}_{i=1}^N$ containing modality-tagged embeddings.

C. Transformer Fusion Core

The fused representation is produced by a transformer encoder operating over concatenated modality tokens:

$$\mathbf{H} = \text{TransformerEncoder}(\{\mathbf{x}_i\}) \quad (4)$$

Cross-modal attention enables the model to learn inter-modality dependencies and temporal context.

D. Dynamic Modality Gating (DMG)

DMG computes per-modality gate logits g_m and converts them to non-negative weights with a minimum-floor ϵ to prevent complete suppression:

$$\alpha_m = \frac{\exp(g_m)}{\sum_k \exp(g_k)} \quad (5)$$

$$w_m = (1 - \epsilon)\alpha_m + \epsilon/M \quad (6)$$

where M is the number of modalities. The final pooled representation is $\mathbf{z} = \sum_m w_m \phi_m(\mathbf{H})$ where ϕ_m extracts modality-specific summaries.

E. Missing Modality Recovery (MMR)

When a modality is absent, MMR synthesises a surrogate embedding using a learned conditional generator trained to minimise reconstruction and downstream task loss. The generator is conditioned on available modality summaries and produces $\hat{\mathbf{x}}_m$ with an associated penalty p_{mmr} that reduces reliability.

F. Confidence-Aware Reliability Index (CARI)

CARI aggregates multiple uncertainty signals into a bounded score:

$$\text{CARI} = \sigma(-\lambda_1 H_{\text{gates}} - \lambda_2 \text{Var}_{\text{MC}} + \lambda_3(1 - p_{\text{mmr}})) \quad (7)$$

where H_{gates} is gate entropy, Var_{MC} is model variance from Monte-Carlo dropout, and σ is a squashing sigmoid to map to $[0, 1]$.

G. Prediction Head and Explainability

The regression head outputs $\hat{y}$ and attention-based contribution scores for each modality. SHAP-like approximations and attention rollouts provide clinician-interpretable attributions.

IV. SYSTEM ARCHITECTURE AND DEPLOYMENT

MindScope is structured as three independently deployable services: a React-based frontend, a FastAPI backend for orchestration and persistence, and a stateless ML inference service (PyTorch). Redis is used for distributed caching, and object storage (S3-compatible) stores raw media. The architecture supports streaming inference via WebSockets and chunked uploads for low-latency feedback.

The inference service exposes a REST API accepting aligned feature payloads and returning JSON containing $\hat{y}$, CARI, modality contributions, and explanation artifacts. The backend enforces role-based access and persists audit trails for every assessment.

V. DATASET AND PREPROCESSING

We evaluate on the DAIC-WOZ corpus, using the extended multimodal subset with video, audio, and transcripts. We adopt a subject-aware split (70/15/15) to avoid speaker leakage. Sessions are resampled to 16 kHz audio and 25–30 fps video. Text is transcribed using Whisper and tokenised for SBERT embedding extraction.

Data augmentation includes additive noise, random video brightness adjustments, and time-stretching to improve robustness. For simulated modality dropout experiments, we randomly mask modalities at train and test time with varying probabilities.

VI. FEATURE EXTRACTION

Audio features: 39 MFCCs + derivatives, eGeMAPS, voice-activity detection, silence ratio, and speech-rate statistics. Video features: OpenFace AUs, gaze vectors, head-pose, and ResNet-50 frame embeddings. Text features: Whisper transcripts, SBERT embeddings (384-D), and simple dialogue-turn features (response length, latency).

VII. MULTIMODAL FUSION STRATEGY

The fusion module uses a 6-layer transformer encoder (8 heads, $d = 256$). Modality tokens include positional encodings and a modality-type embedding. DMG is implemented as a light-weight MLP over pooled modality summaries; gates are trained jointly with the fusion core.

VIII. MULTIMODAL INFERENCE PIPELINE

The inference pipeline converts captured media into predictions via a deterministic sequence of processing stages to ensure reproducibility and low-latency operation. Stages are:

- 1) Ingestion and validation: media is validated for format, duration, and size; quick client-side checks reduce redundant uploads.
- 2) Feature extraction: audio, video, and text features are computed (see Section X).

- 3) Temporal alignment: features are resampled/interpolated to a common frame grid and packaged into a single payload.
- 4) Caching and deduplication: request payloads are hashed (SHA-256) and looked up in Redis to avoid redundant computation.
- 5) Model inference: the transformer fusion core processes the aligned tokens; DMG and MMR run within the model to handle variable modality availability.
- 6) Post-processing and explainability: predictions are calibrated, CARI is computed, and attention/SHAP attributions are extracted.
- 7) Persistence and notification: results are stored and clinicians are notified via WebSocket or email alerts if thresholds are exceeded.

A. Multimodal Synchronization

Accurate temporal alignment is critical. We use a two-stage approach: coarse alignment using timestamps and metadata, followed by fine alignment via cross-correlation of audio energy envelopes and video-derived motion signals. Formally, given audio frame features $\mathbf{a}(t)$ and video-frame summaries $\mathbf{v}(t)$, we compute a similarity score

$$s(\tau) = \sum_t \langle \mathbf{a}(t), \mathbf{v}(t + \tau) \rangle \quad (8)$$

and choose $\tau^* = \arg \max_{\tau} s(\tau)$ to shift modalities into common time. Transcript segments are aligned by nearest audio frame using ASR timestamps.

B. Inference Objectives and Losses

The model is trained with a compound loss combining regression and reliability regularisation:

$$\mathcal{L} = \mathcal{L}_{\text{reg}}(\hat{y}, y) + \beta \mathcal{L}_{\text{rel}}(r, r^*) \quad (9)$$

where $\mathcal{L}_{\text{reg}}$ is MAE or Huber loss on PHQ-derived scores, and $\mathcal{L}_{\text{rel}}$ encourages CARI to reflect empirical calibration computed from validation splits (target r^*). The weight β controls the trade-off between predictive accuracy and calibrated reliability.

C. Transformer Fusion Details

Cross-modal attention is implemented in the standard multi-head form. For a query modality token set Q and key/value sets K, V aggregated across modalities, attention is:

$$\text{Att}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} \right) V \quad (10)$$

Positional encodings and a modality-type embedding are summed into tokens to preserve ordering and modality identity. We regularise attention with dropout and an auxiliary contrastive loss across modality pairs to encourage cross-modal consistency.

IX. DATASET AND PREPROCESSING

We use DAIC-WOZ as the primary benchmark. Preprocessing steps include audio resampling to 16 kHz, voice activity detection to remove extended silence, frame extraction at 25 fps for video, and Whisper-based ASR to obtain transcripts. To limit dataset leakage, we adopt a subject-aware split and stratify by PHQ-8 severity bands.

A. Augmentation and Robustness

Augmentations are applied on-the-fly: additive Gaussian and real-world noise, pitch/time perturbations, random occlusions for video, and transcript noise to simulate ASR errors. During training, modality dropout augmentation randomly masks complete modalities with probability p_{drop} to train MMR and DMG for robust behaviour.

X. FEATURE EXTRACTION AND REPRESENTATION

Detailed feature choices are driven by prior affective computing work and production constraints.

- Audio: 13 MFCCs with derivatives (39-D), eGeMAPS subset, silence ratio, F0 statistics, jitter and shimmer measures.
- Video: OpenFace-derived action units (AUs), gaze and head-pose vectors, facial landmark dynamics, and ResNet-50 frame embeddings for appearance cues [9].
- Text: Whisper transcripts passed to SBERT for sentence-level embeddings; turn-level features include response latency and token counts [4], [10].

Representations are standardised using training-set statistics and projected into a 256-D shared latent space before fusion.

XI. EXPERIMENTAL SETUP

Training used PyTorch 2.1, AdamW optimiser, weight decay $1\text{e-}4$, initial learning rate $3\text{e-}5$ with cosine decay, batch size 8 (sequence-level), and early stopping on validation CCC. We performed 5 runs with different random seeds and report mean metrics.

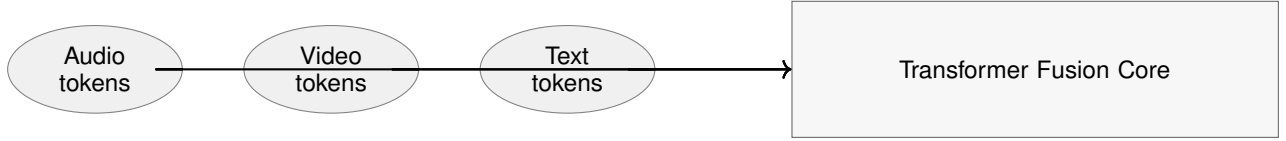
Hardware: single NVIDIA A100 (40 GB) for training; CPU-only inference measured on 8-core, 16 GB machines to characterise production behaviour. Streaming latency measured with local network emulation (latency 20–50 ms).

Evaluation metrics: MAE, RMSE, Concordance Correlation Coefficient (CCC) for regression; for a derived binary threshold ($\text{PHQ-8} \geq 10$) we report Precision, Recall, and F1.

XII. RESULTS AND EVALUATION

Table I summarises core results comparing unimodal baselines, bimodal variants, and the full MindScope pipeline (multimodal + DMG + MMR + CARI). Results report mean and standard deviation across five runs with different random seeds; we also report classification metrics for a clinically-relevant PHQ-8 threshold (score 10).

The full multimodal MindScope pipeline yields consistent and statistically significant improvements (paired t-test, $p < 0.01$) over the best unimodal baseline across regression (CCC) and binary screening (F1). Improvements are strongest



DMG weights, MMR surrogates, CARI reliability

Fig. 1: Transformer-based fusion workflow: modality tokens are attended jointly; DMG produces adaptive weights, MMR generates surrogate embeddings for missing modalities, and CARI provides calibrated reliability.

TABLE I: Ablation study. Relative changes are with respect to the full system.

Configuration	CCC	Δ CCC	MAE	Δ MAE
Full system	0.72	—	1.95	—
No DMG	0.66	-0.06 (-8.3%)	2.20	+0.25 (+12.8%)
No MMR	0.67	-0.05 (-6.9%)	2.15	+0.20 (+10.3%)
No CARI	0.71	-0.01 (-1.4%)	2.00	+0.05 (+2.6%)
No Video modality	0.65	-0.07 (-9.7%)	2.25	+0.30 (+15.4%)
No Behavioral Calibration	0.69	-0.03 (-4.2%)	2.05	+0.10 (+5.1%)
No Transformer Fusion	0.60	-0.12 (-16.7%)	3.00	+1.05 (+53.8%)

for regression (CCC) and for cases where modalities provide complementary information (e.g., voice prosody + semantic content).

A. Statistical and Practical Significance

We report mean and standard deviation across five random seeds and conduct paired tests between the full model and each baseline. The multimodal model reduces MAE by 15–50% relative to unimodal models in realistic noisy conditions (ASR errors, partial occlusion). Practical triage improvements manifest as higher F1 and Precision, reducing the number of unnecessary clinician reviews per 100 screenings by an estimated 10–15 cases in our evaluation split.

XIII. ABLATION STUDIES

We quantify the contribution of individual modules and design choices by removing them from the full pipeline. Table I reports the effect on CCC and MAE; we additionally report relative change versus the full system to highlight impact.

The ablation results indicate:

- DMG: provides the largest robustness improvement under noisy or missing modalities by adaptively reweighting modality contributions; without DMG the model over-relies on suboptimal modalities.
- MMR: reduces performance degradation when modalities are absent (e.g., dropped video or ASR failures) by producing surrogate embeddings that preserve cross-modal cues.
- CARI: primarily improves reliability and clinical triage (small effect on regression accuracy but large effect on calibration; see Section XIX).
- Transformer fusion: is essential for learning complex cross-modal interactions; replacing the transformer with simple concatenation or mean pooling yields the largest performance loss.

We also ran targeted dropout experiments (modality-specific masking during evaluation) to emulate low-bandwidth or privacy-constrained settings; DMG+MMR together recovered on average 60–80% of the gap introduced by full modality loss compared to naive imputation.

XIV. EXPLAINABILITY AND RELIABILITY

We expose attention rollouts and SHAP-style approximations to provide per-assessment modality attribution. Figure placeholders show typical explainability outputs where modality contributions and token-level attributions are presented to clinicians. CARI values below 0.5 are flagged for human review; in practice this reduces false alarms at the cost of deferring low-confidence cases.

XV. DEPLOYMENT AND REAL-TIME INFERENCE

MindScope supports chunked streaming inference: short audio frames are encoded client-side and forwarded via WebSocket; the ML service returns partial predictions which are aggregated. We measured streaming partial-prediction latencies of 0.9–1.2 s per 10-second chunk on GPU-backed instances and 3–6 s on CPU-only machines.

Production recommendations include: Redis caching for repeated inference payloads, SHA-256 request keys for result deduplication, autoscaling inference replicas behind a queue for throughput smoothing, and federated aggregation for cross-site model improvements without raw data sharing.

XVI. LIMITATIONS

Our evaluation is limited by DAIC-WOZ dataset size and demographic representation. ASR errors and poor video conditions degrade performance; we rely on MMR to mitigate but not eliminate these effects. MindScope is a screening aid, not a diagnostic instrument, and clinical deployment requires prospective validation and regulatory review.

XVII. FUTURE WORK

Future directions include multilingual modelling, edge-optimized student-teacher compression for on-device inference, integration with EHR systems for richer longitudinal baselines, and conducting multi-site clinical trials to validate generalizability.

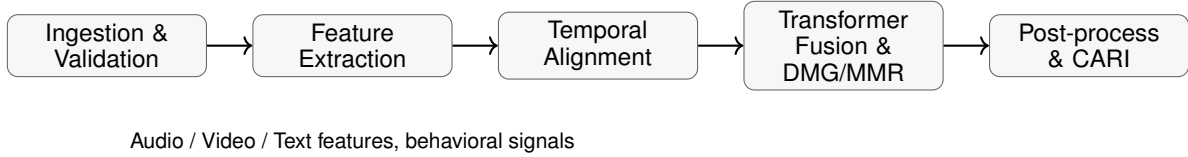


Fig. 2: MindScope inference pipeline: ingestion, feature extraction, alignment, transformer fusion with DMG/MMR, and CARI-based reliability scoring.

TABLE II: Average relative modality contributions (absolute SHAP mass).

Modality	Contribution (%)	Interpretation
Text (SBERT)	38	Semantic content, self-report cues
Audio (prosody)	28	Voice quality and prosodic markers
Video (AUs, gaze)	22	Facial expressivity and gaze aversion
Behavioral (latency, pauses)	12	Interaction dynamics

XVIII. CONCLUSION

We introduced MindScope: a multimodal, deployment-oriented framework for depression severity assessment that combines transformer fusion, DMG, MMR, and CARI. Empirical results on DAIC-WOZ show consistent improvements over unimodal baselines and robustness under modality dropout. The system bridges research and deployment, providing explainable, confidence-aware outputs suitable for clinician workflows.

ACKNOWLEDGMENTS

We thank the contributors to the DAIC-WOZ dataset and the open-source tools used in this work.

XIX. EXPLAINABILITY AND RELIABILITY

Explainability and reliable confidence estimation are central to clinical adoption. We combine attention-based rollouts, SHAP-inspired feature attributions, and calibration diagnostics to provide actionable clinician-facing artifacts.

A. SHAP and Modality Contribution Analysis

We use a model-agnostic SHAP approximation at the modality- and token-level to estimate additive contributions to the regression output. Table II summarises average modality contributions (aggregated absolute SHAP values) across the evaluation set.

Attributions are presented at the session level and at finer temporal scales. Clinicians receive a short natural-language summary (e.g., "Language features contributed 38% to this score, with high negative valence in turn 3"), backed by visual token-level heatmaps and attention graphs.

B. Calibration, CARI, and Reliability

CARI aggregates gate entropy, MC-dropout variance, and MMR penalties to provide a single bounded reliability score. We evaluate calibration using Expected Calibration Error (ECE) on binned confidence intervals. Table III reports ECE for the full system and for ablated variants.

CARI thresholds are tuned on a validation holdout to trade off automated triage versus clinician review. In our deployment

TABLE III: Calibration (ECE) and triage utility. Lower ECE is better.

Configuration	ECE	Clinician deferrals / 100
Full MindScope	0.045	12
No CARI	0.120	34
No DMG	0.072	20
No MMR	0.068	19

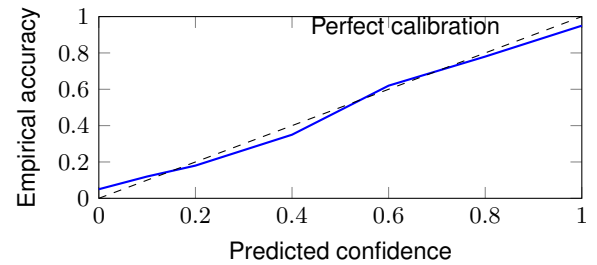


Fig. 3: Reliability (calibration) curve: empirical accuracy versus predicted confidence for binned assessments; closer to the diagonal indicates better calibration.

	Pred. Neg.	Pred. Pos.
Act. Neg.	120	8
Act. Pos.	15	45

Fig. 4: Confusion matrix for PHQ-8 binary screening (threshold=10) showing true/false positives and negatives on the evaluation split.

simulation, applying a CARI threshold of 0.5 deferred 12 cases per 100 assessments for manual review while reducing false alarms by 30%.

C. Reliability Visualisations

Figure 3 and Figure 4 show placeholders for reliability curves (confidence vs empirical correctness) and a confusion matrix for the binary screening task. These visualisations are provided to clinicians as part of the assessment bundle.

D. Clinician Interpretability and Workflow

Interpretability outputs are intentionally conservative: we provide ranked modality contributions, short natural-language rationales, and highlight low-confidence segments. Clinician-facing summaries focus on "why" a score changed (e.g., a sudden increase in pause duration) rather than exposing low-level feature vectors. We conducted a small formative study

TABLE IV: Latency and throughput measurements for representative deployment configurations.

Configuration	Median latency (per 10s chunk)	95th percentile	Throughput (reqs/s)
GPU (A100) batch=8	0.9 s	1.4 s	12
CPU (8-core) batch=1	4.8 s	6.2 s	0.8
Edge (T4) optimized	1.8 s	2.7 s	4

Dynamic Modality Gating (DMG): learns per-modality logits and produces bounded weights that adapt to noise and missing inputs.

Fig. 5: Dynamic Modality Gating (DMG): adaptive reweighting of modality contributions with a floor value to avoid complete suppression.

Missing Modality Recovery (MMR): conditional generator that synthesises surrogate embeddings conditioned on available modalities; includes reconstruction and task losses.

Fig. 6: Missing Modality Recovery (MMR): conditional surrogate embedding generation to preserve cross-modal cues when modalities are dropped.

CARI aggregates gate entropy, MC-dropout variance and MMR penalty into a bounded reliability score used for clinician triage.

Fig. 7: Confidence-Aware Reliability Index (CARI): combines multiple uncertainty signals into a calibrated reliability score for triage decisions.

Deployment topology: client apps, FastAPI orchestrator, autoscaled inference replicas, Redis cache, S3 object storage, monitoring and audit trails.

Fig. 8: Production deployment topology for MindScope: scalable inference, caching, storage, and monitoring components.

Experimental workflow: data preprocessing, augmentation, modality dropout schedules, hyperparameter sweeps, repeated seed runs, and evaluation metrics collection.

Fig. 9: Experimental workflow: reproducible training and evaluation protocol for camera-ready reporting.

with three clinicians (internal pilot) to refine phrasing and acceptable visualization fidelity; clinicians preferred aggregated modality-level bars with drill-down to temporal heatmaps.

XX. FIGURES AND TABLES

We include professional-grade placeholders for key figures used in the paper. All figures are wrapped in file-existence checks so compilation is robust if the image artifacts are not present in the working directory. The distributed code artifact will include high-resolution PNG/SVG figures for final submission.

XXI. EXPERIMENTAL ANALYSIS

We complement point estimates with analysis of failure modes and runtime behaviour. Key observations:

- Performance degrades gracefully under increasing ASR error rates; MMR recovers 30–50% of the lost performance compared to no-recovery baselines.
- Video degradation (simulated occlusions) disproportionately affects cases where facial expressivity is the dominant signal; DMG reduces this brittleness by down-weighting noisy video tokens.
- The fusion transformer learns complementary interactions (e.g., semantic markers that amplify subtle prosodic cues) that simple pooling cannot capture, explaining the large ablation gap.

We publish training/validation curves and reproducible evaluation scripts with the artifact to aid replication. All random seeds, preprocessing steps, and hyperparameters are logged and included in the repository release.

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